



Drug Interactions

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Drug interactions are changes in a drug’s effects due to recent or concurrent use of another drug(s) (drug-drug interactions), ingestion of food ([drug-nutrient interactions](#)), or ingestion of dietary supplements ([dietary supplement-drug interactions](#)).

A drug-drug interaction may increase or decrease the effects of one or both drugs. Clinically significant interactions are often predictable and usually undesired (see table [Some Medications With Potentially Serious Drug-Drug Interactions](#)). Adverse effects or therapeutic failure may result. Rarely, clinicians can use predictable drug-drug interactions to produce a desired therapeutic effect. For example, coadministration of lopinavir and [ritonavir](#) to patients with HIV infection results in altered metabolism of lopinavir and increases serum lopinavir concentrations and effectiveness.

TABLE	
Some Medications With Potentially Serious Drug-Drug Interactions*	
Mechanism	Examples
Narrow margin of safety†	Antiarrhythmic drugs (eg, quinidine)
	Antineoplastic drugs (eg, methotrexate)
	Digoxin
	Lithium
	Theophylline
Extensive metabolism by certain hepatic enzymes	Warfarin
	Alprazolam
	Amitriptyline
	Atorvastatin
	Carbamazepine

Mechanism	Examples
	Clozapine Corticosteroids Cyclosporine Diazepam HIV protease inhibitors Imipramine Lovastatin Midazolam Olanzapine Phenytoin Sildenafil Simvastatin Tacrolimus Tadalafil Theophylline Triazolam Vardenafil Warfarin
Inhibition of certain hepatic enzymes‡	Aprepitant Boceprevir Cimetidine Ciprofloxacin Clarithromycin Cobicistat Conivaptan Diltiazem Erythromycin Fluconazole Fluoxetine

Mechanism	Examples
	Fluvoxamine Itraconazole Ketoconazole Paroxetine Posaconazole Ritonavir Telaprevir Telithromycin Verapamil Voriconazole
Induction of certain hepatic enzymes	Barbiturates (eg, phenobarbital) Bosentan Carbamazepine Efavirenz Phenytoin Rifabutin Rifampin St. John's wort
* Any medication to be used concurrently with one of these medications should be thoroughly evaluated for possible interactions. † Even when used alone, these medications may have serious adverse effects. Concurrent use of another medication that increases the action of these medications further increases risk of adverse effects. For additional research on potential drug-drug interactions, consult a reliable source, such as the Drugs.com Drug Interaction Checker. ‡ Inhibition also can occur after ingestion of grapefruit products.	

In therapeutic duplication, 2 medications with similar properties are taken at the same time and have additive effects. For example, taking a benzodiazepine for anxiety and another benzodiazepine at bedtime for insomnia may have a cumulative effect, leading to toxicity.

Drug interactions involve

- [Pharmacodynamics](#)
- [Pharmacokinetics](#)

In **pharmacodynamic interactions**, one medication alters the sensitivity or responsiveness of tissues to another medication by having the same (agonistic) or a blocking (antagonistic) effect. These effects usually occur at the receptor level but may occur intracellularly.

In **pharmacokinetic interactions**, a medication alters absorption, distribution, protein binding, metabolism, or excretion of another medication. Thus, the amount and persistence of available medication at receptor sites is changed. Pharmacokinetic interactions alter magnitude and duration, not type, of effect. They can often be predicted based on knowledge of the individual medications or detected by monitoring medication concentrations or clinical signs.

Minimizing drug interactions

Clinicians should know all of their patients' current medications, including medications prescribed by other clinicians and all over-the-counter medications, herbal products, and nutritional supplements. Asking patients relevant questions about diet and alcohol consumption is recommended. The fewest medications in the lowest doses for the shortest possible time should be prescribed. The effects, desired and undesired, of all medications taken should be determined because these effects usually include the spectrum of medication interactions. If possible, medications with a wide safety margin should be used so that any unforeseen interactions do not cause toxicity.

Patients should be observed and monitored for adverse effects, particularly after a change in treatment; some interactions (eg, effects that are influenced by enzyme induction) may take ≥ 1 week to appear. Drug interactions should be considered as a possible cause of any unexpected problems. When unexpected clinical responses occur, prescribers should determine serum concentrations of selected medications being taken, consult the literature or an expert in drug interactions, and adjust the dosage until the desired effect is produced. If dosage adjustment is ineffective, the medication should be replaced by one that does not interact with other medications being taken.

Drug Information for the Topic



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